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Effects of Fin Static Aeroelasticity on Roll Rate 05 October 2013

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Introduction

Thin fins are commonly flown on small sounding rockets. The airfoils used can be simple Al plate or they can be built up by wrapping carbon fiber around a foam core. This memo addresses the effects of subsonic static aeroelastic twisting on roll rate. The motivation is the roll rate profile for CSULB's P-18 D rocket as shown in Fig. 1. Here the roll rate starts to accelerate a few seconds prior to the failure at pitch-roll resonance. The question that motivated this study is whether or not this increase was due to fin static aeroelasticity. This memo documents the Excel™ code, TWISTIT.xls used to answer this question.

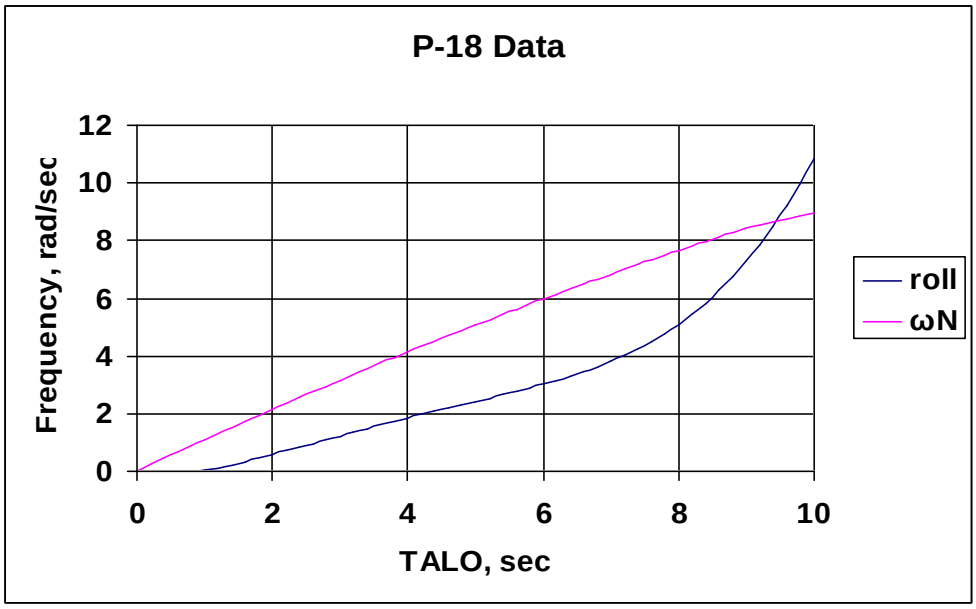


Figure 1 P-18 Flight test data

Nomenclature

Mnemonic	Definition
A, B	Parameters in the fin chord vs. spanwise coordinate relation
b	Fin semi span, ft from body centerline
C_L	Rolling moment coefficient $=\frac{L}{2qSR}$
C_{Lp}	Roll damping moment coefficient $=\frac{\partial C_L}{\partial(pR/U)}$

Mnemonic	Definition

$C_{L\delta}$	Roll driving moment coefficient $=\frac{\partial C_L}{\partial \delta}$
$c_{N\alpha}$	Fin average airfoil normal force coefficient slope, rad^{-1}
C_1, C_2	Constants of integration
c	Local fin chord, ft
c_R	Root chord, ft
c_T	Tip chord, ft
G	Torsional modulus of elasticity of the fin material, lb/ft^2
J	Torsional stiffness of a fin airfoil, ft^4
L	Rolling moment, ft-lb
N	Number of fin panels
p	Roll rate, rad/sec
A, P, D	Functions in the torsional solutions
q	Dynamic pressure, lb/ft^2
R	Body radius, ft
S	Aerodynamic reference area, ft^2
T	Torque about the elastic axis, ft-lb
t	Total fin thickness, ft
U	Flight speed, ft/sec
w	Fin skin thickness, ft
y	Distance from center line to a fin strip, ft
α	Local angle of attack, rad
δ	Cant angle of a single fin panel, rad
θ	Torsional deflection (twist angle), rad

Analysis

First, the assumptions: The fin planform is assumed to be a clipped delta as sketched. Next, the Mach number is subsonic. The phenomena discussed here tend to be much less noticeable at supersonic speeds because there the airfoil center of pressure and elastic axis are much closer. All angles of attack are small compared to unity. The fin airfoil shape can be modeled as a flat plate. As is customary, the roll moment of inertia is negligibly small.

The simplified aerodynamic model used here is strip theory. In this approach the local normal force coefficient slope is approximated by an average taken over the entire planform. Considering the very approximate nature of the elastic twisting model employed, it does not seem worthwhile to expend much effort on refining the airload distribution.

It is common to assert that designers choose fin sweep back to reduce fin wave drag or to increase critical Mach number. In truth, it must be chosen to mitigate the possibility of a static aeroelastic failure due to chordwise curling of the leading edge in the tip region. It is assumed that the fin is swept back sufficiently to preclude chordwise bending; it can therefore be neglected. Swept fins are also known for coupling between torsion and spanwise bending. It is assumed that static bending-torsion coupling can be neglected because the coupling effects of fin cant and roll damping are approximately equal, but of opposite sign.

The elastic model used considers the fin to be a slender beam of high aspect ratio. Results reported in ref.(2) indicate that the slender beam approximation will overestimate the torsion deflection by about 20%. However, all we are trying to do is provide the designer with a tool that indicates trends. For that purpose this error is acceptable.

Figure 2 below displays the fin geometry:

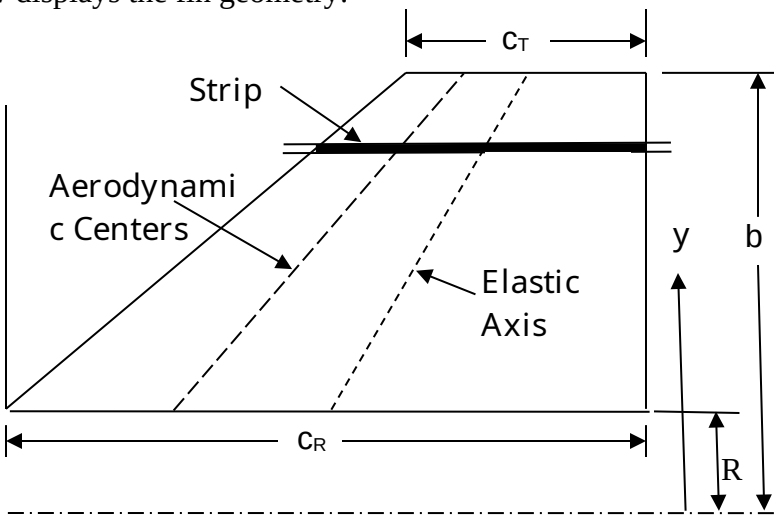


Figure 2 Fin Geometry

Torsional Deflections

Consider the fin to be a one-dimensional beam subject to torsional deformations. Assume the elastic axis lies on the fin half chord line. Since subsonic flow has been assumed, the aerodynamic loading will be centered on the quarter chord line as sketched in Fig. 2. The air loads of interest are due to torsion, cant angle, roll damping and angle of attack. Then, the torque about the elastic axis from a strip of fin will be:

$$dT = qcdyc_{N\alpha}(\theta + \alpha(1 + \frac{R}{y^2}) + \delta - \frac{py}{U}) * c/4 \qquad (1)$$

The change in twist angle over the strip is given by the St. Venant theory

$$dT = GJ\, d/dy(d\theta/dy)\, dy = GJ \frac{d\theta'}{dy} dy \qquad (2)$$

Torsional Stiffness

There are two kinds of fin structure to be considered: First, a solid plate[1] for which

$$J = 1/3ct^3. \qquad (3a)$$

[1] A more accurate approximation is $J = 1/3ct^3(1 - 0.63t/c)$. In most practical cases, the two formulae differ by only a few percent.

Second, a built-up structure for which thin skins of thickness w are laid up on a foam or balsa core to form a flat plate airfoil. Neglecting the contribution of the core leads to Bredt's formula for the torsional stiffness:

$$J = 2c(t - w)^2 t \tag{3b}$$

Note that the chord varies with the spanwise coordinate according to:

$$c = \frac{c_R b - c_T R}{b - R} - \frac{c_R - c_T}{b - R} y = A - By \tag{4}$$

Putting these pieces together gives

$$\frac{d^2\theta}{dy^2} - \frac{qc_{N\alpha}}{4G(J/c)} c\theta = \frac{qc_{N\alpha}}{4G(J/c)} c\left[\alpha\left(1 + \frac{R^2}{y^2}\right) + \delta - \frac{py}{U}\right] \tag{5a}$$

For a straight-tapered planform, we can change the independent variable from y to c:

$$\frac{d^2\theta}{dc^2} - Kc\theta = Kc\left[\delta - \frac{py}{U}\right] \tag{5b}$$

where

$$K = \frac{qc_{N\alpha}}{4G(J/c)}\left(\frac{b - R}{c_R - c_T}\right)^2 \tag{5c}$$

The initial conditions for a root chord clamped to the body are:

$$\left.\begin{aligned} \theta = 0, \text{ and} \quad & \text{(6a)} \\ \frac{d\theta}{dy} = \frac{d\theta}{dc} = 0 \quad & \text{(6b)} \end{aligned}\right\} @\ y = R \text{, or } c = c_R$$

Equations (5) have a general solution in terms of modified Bessel functions of the first kind and of fractional order. However Excel™ only supports I functions whose order is a positive integer. Therefore, recourse to numerical methods will be taken.

Because the θ response is proportional to the appropriate right hand side, each of the three θ solutions takes the form

$$\begin{aligned} \theta(\delta, c) &= \delta D(c) \\ \theta(p/U, c) &= (p/U) P(c) \\ \theta(\alpha, c) &= \alpha A(c) \end{aligned} \tag{8}$$

Then, the functions A, D and P can be found numerically integrating:

$$\begin{aligned}
\frac{d^2D}{dc^2} - KcD &= Kc \\
\frac{d^2P}{dc^2} - KcP &= -Kc\left(\frac{A-c}{B}\right) \\
\frac{d^2A}{dc^2} - KcA &= Kc\left(1+\frac{R^2}{y^2}\right)
\end{aligned} \tag{9}$$

subject to the initial conditions (eq's. (6))

Loads Acting on a Flexible Fin

Now, suppose we knew the three solutions, $\theta(\delta, y)$, $\theta(p/U, y)$ and $\theta(\alpha, y)$. We next show how to find the "elasticized" loading. Consider the angle of attack response on a strip of span dy. The rigid fin C_{Lp} remains the same as always. But, aeroelastic effects increase C_{Lp} by

$$\Delta C_{Lp} = \frac{c_{N\alpha}}{2R^2S}\int_R^b c(y)yP(c)dy,$$

(10a)

and the corresponding increment in $C_{L\delta}$ is

$$\Delta C_{L\delta} = \frac{c_{N\alpha}}{2RS}\int_R^b c(y)yD(c)dy.$$

(10b)

Futhermore, there is an change to the normal force, including upwash around the body, acting on a fin panel

$$\Delta C_{N\alpha} = \frac{c_{N\alpha}}{S}\int_R^b c(y)A(y)dy.$$

(10c)

Thus, integrate eq's. (9) , and at the same time, evaluate eq's. (10) as auxiliary integrals. Use eq. (4) to relate c and y.

Roll Equilibrium

Roll equilibrium is achieved when the two roll torques are in balance. First, recapitulate the rigid fin strip theory for $C_{L\delta}$ and C_{Lp}. Consider a strip of span dy for which

d(Roll damping moment) $= qcdyyc_{N\alpha} * \frac{-py}{U} = -qcy^2 c_{N\alpha} \frac{p}{U} dy$.

Then, roll damping moment

$$= -qc_{N\alpha} \frac{p}{U} \int_R^b (A - By) y^2 dy = -qc_{N\alpha} \frac{p}{U} \left[A(b^3 - R^3)/3 - B(b^4 - R^4)/4 \right]$$

Then, for one fin panel

$$C_{Lp} = -\frac{c_{N\alpha}}{2R^2 S} \left[A(b^3 - R^3)/3 - B(b^4 - R^4)/4 \right]$$

(11a)

d(Roll driving moment) $= qcdyyc_{N\alpha}\delta$. And, for one fin panel,

$$C_{L\delta} = \frac{c_{N\alpha}}{2RS} \left[A(b^2 - R^2)/2 - B(b^3 - R^3)/3 \right] \qquad (11b)$$

The full stability derivatives are the sum of eq's. (10) and (11). The equilibrium spin rate is estimated by setting sum of all the roll moments equal to zero, and solving for the roll rate. Then, if all fin panels have the same cant angle,

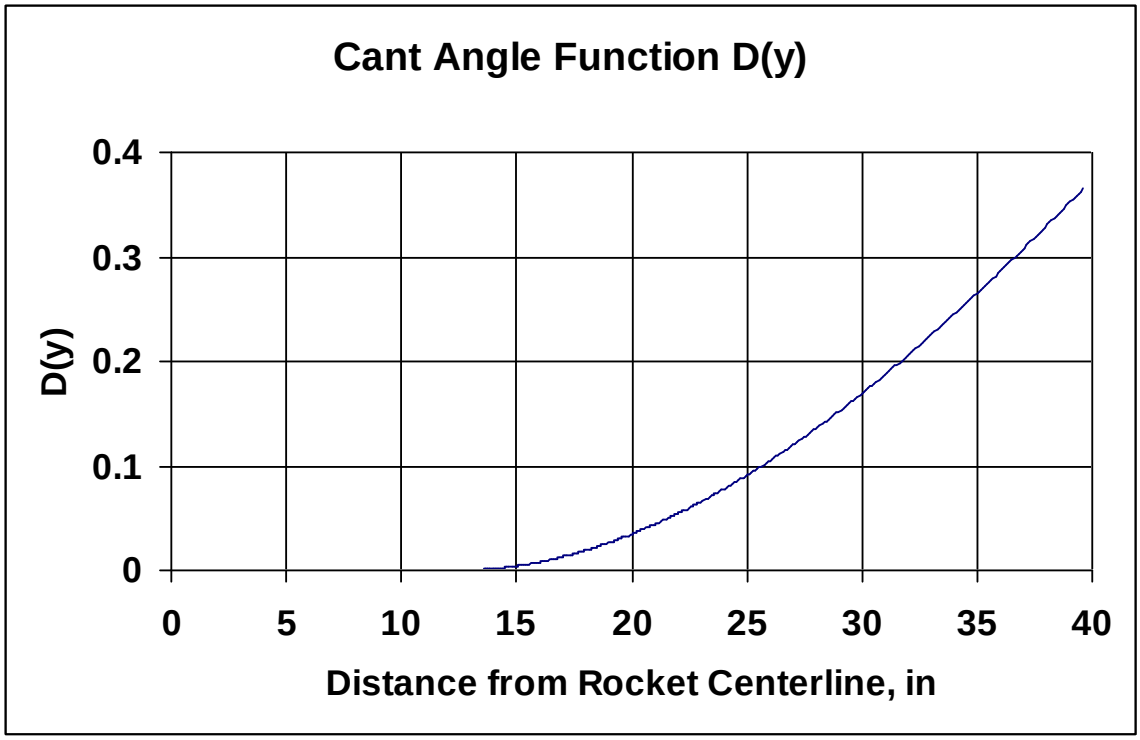
$$C_{L\delta}\delta = C_{Lp} \frac{pR}{U} \qquad (12)$$

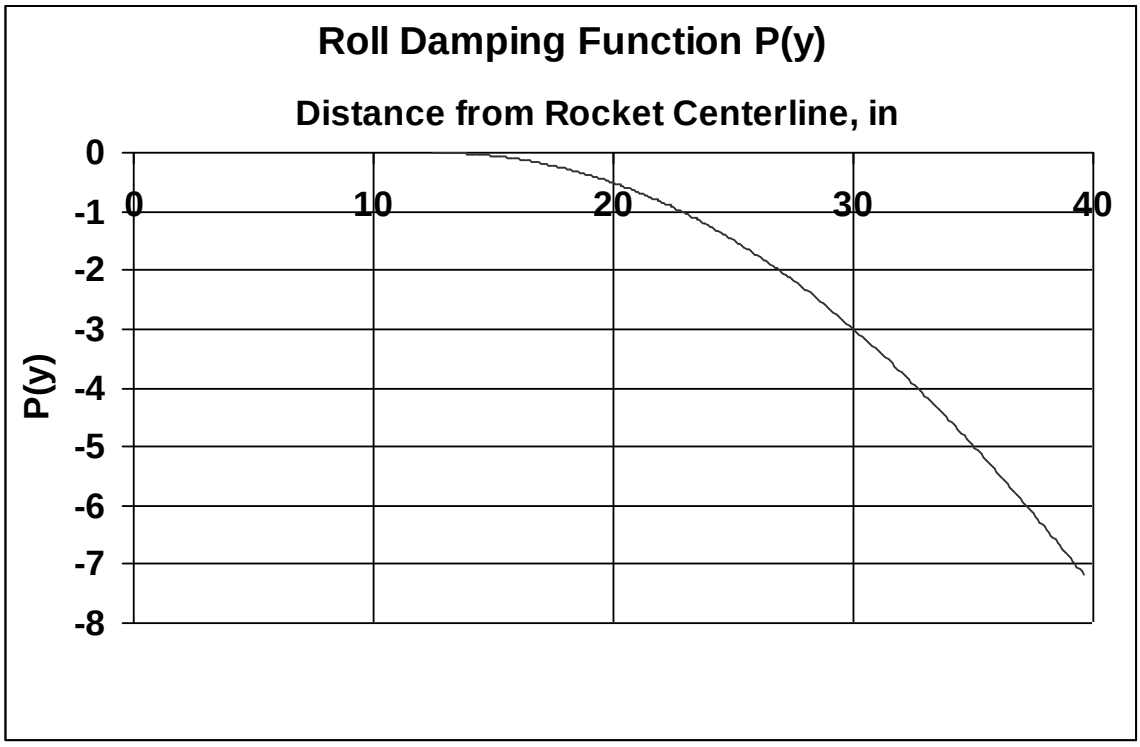
Equation (12), with its aeroelastic corrections, can be used to predict roll rate given fin cant angles, or vice versa.

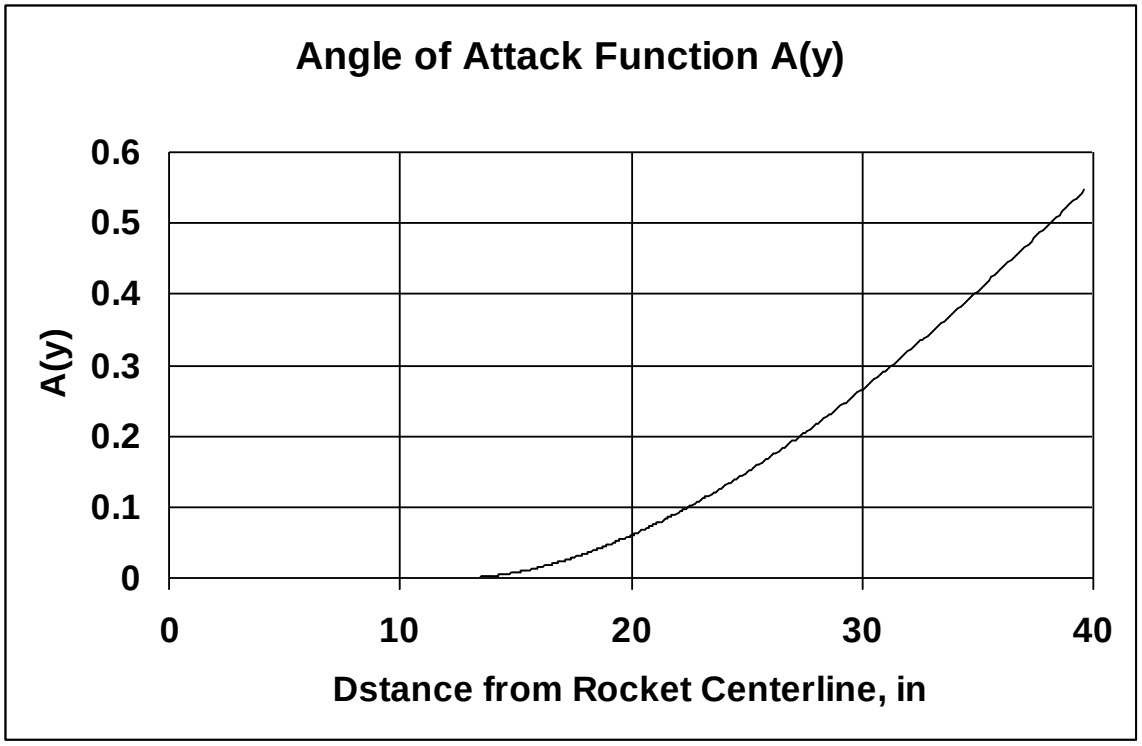
Results

The model described above has been coded into Excel™ and filed as TWISTIT.xls. For typical numbers, data for P-18D was used, and the results for D, P and A plotted below:

Parameter	Value
Root chord	24"
Tip chord	4"
Body Radius	12.5"
Fin semispan from body centerline	39.6"
Fin total thickness	0.375"
Fin skin thickness	0.02742"
Fin J/c	0.00771 in^3
Dynamic pressure	800 psf


Cant Angle Function D(y)
0.4
0.3
0.2
0.1
0
D(y)
0
5
10
15
20
25
30
35
40
Distance from Rocket Centerline, in

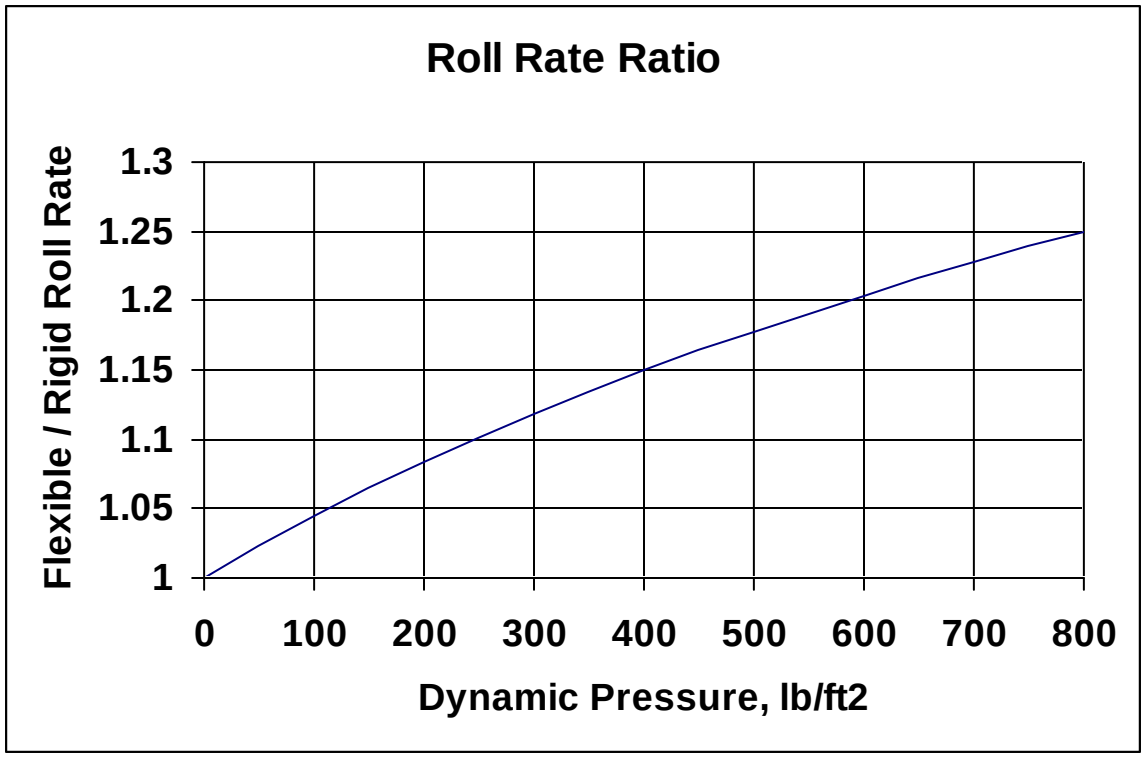

Roll Damping Function P(y)
Distance from Rocket Centerline, in
0
-1
-2
-3
-4
-5
-6
-7
-8
P(y)
0
10
20
30
40



The following table summarizes the results:

Parameter	$C_{L\delta}$	C_{Lp}
Rigid	2.399	-4.808
Flexible increment	4.426	-6.141
Total	6.825	-10.949

We immediately discover that the flexible fin roll rate is 25% greater than its rigid counterpart. The general dependence of the flexible roll rate to rigid roll rate ratio on dynamic pressure is shown below. Note that the maximum dynamic pressure observed during the P-18D flight was only 673 lb/ft^2.



Summary

This very approximate analysis shows that fin static aeroelasticity effects probably operated to increase the P-18D roll rate beyond its rigid fin value, but most likely there is an additional cause to the increase in roll rate observed near failure.

1. F. B. Hildebrand, "Advanced Calculus for Engineers", Prentice-Hall, Inc., New York, 1949
2. R. L. Bisplinghoff, H. Ashley and R. L. Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, Cambridge, 1956